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Assignment 1
Cognitive Models
2024 Spring
Due by 11:59pm March 30

AI USE STATEMENT: I did not use AI tools on the content of this assignment.

Question 1: Reflecting on Poggio's (2012)'s main points and consider whether the 'level of learning' should be incorporated as a fourth level of analysis in Reinforcement Learning, as explored by Niv and Langdon (2017) (Would it be strange to add a level of learning to a task that is already defined by learning?!). Discuss your reasoning, and elucidate how it aligns or does not align with Poggio's key arguments (30%).

Poggio, a close collaborator of Marr in his development of Marr's three levels of analysis (computational, algorithmic/representational, and implementational), proposes an updated understanding of Marr's 3 levels and proposes the addition of a fourth level, learning. Poggio first emphasizes that there exists connections between the three existing levels of analysis, although Marr talked about their distinctness. Poggio then goes on to point out that in Marr's *Vision,* Marr doesn't really explore intelligence in relation to the brain. Poggio believes that understanding how organisms evolve and learn in their environments, arguably the core idea behind intelligence, would have been extremely relevant to Marr's discussion of the process of vision. It seems as though Poggio proposes the addition of learning to specifically account for the phenomenon of intelligence in the brain. While I agree, that human learning seems to be a neural process that the other three levels of analysis don't necessarily fully account for, Marr's levels of analysis serve to understand information processing systems; learning may be a necessary level to add for understanding intelligent systems, like human brains, but it is not a necessary level to understand all information processing systems. If we see computer run reinforcement learning as an intelligent system, then it may be helpful to apply the learning level of analysis to our understanding of it, but at this point in time, reinforcement learning can be understood as an information processing system.

It may be confusing to say that the brain, an information processing system to many, should also be evaluated on the learning level, but other information processing systems should not be. I say this because while the brain can be understood as an information processing system, when we apply Marr's levels of analysis to cognitive and neural processes, we are not discussing the brain in its totality. The brain and our minds are information processing systems, but so much more. A record player, for example, is an information processing system, but we do not call a record player a brain. To make learning a level of analysis necessary to understand information processing systems would disqualify many machines as information processing systems. Although Poggio is the one proposing this extra level, I don't believe he would disagree with the points I'm making. Evaluating Marr's theory of vision Poggio says, "It should also address how objects are learned for recognition, and describe how circuits and cells learn from visual experience during development…To address so many different levels of understanding and how they interact is a tall order, but it is a necessary prerequisite for any serious theory of intelligence"(Poggio, 1019). Here, Poggio references intelligence, which I qualify as *human* intelligence, since the only intelligence most of us are sure exists is that of humans. Describing learning seems necessary to understanding intelligence as it relates to our cognitive processes and abilities but to apply that to reinforcement learning, we must have already accepted that reinforcement learning is a type of intelligence (which I believe is outside the scope of my knowledge and this paper).

In addition, Niv and Langdon, quite convincingly, lay out an understanding of reinforcement learning with the existing three levels of analysis. In fact, they hint at the limitation of the learning level of analysis: "Although RL applications often use a predesigned representation, in the animal (and human kingdom most task representations are not 'given' but rather must be learned through experience, raising the algorithmic question of how this computational goal can be achieved" (Niv & Langdon, 3). Poggio does something similar: "How then does statistical learning theory, as developed so far, compare with brains? One of the most obvious differences is the ability of people and animals to learn from very few examples" (Poggio, 1019-1020). They are pointing out a fundamental difference between animals, humans, and machine reinforcement learning. The learning which a machine performing reinforcement learning does is different from those of animals and humans. While the computational analysis of human and animal reinforcement learning could benefit from bringing learning back into the discussion (which Niv and Langdon discuss later in the computational analysis section), the computational level goal for machine reinforcement learning is definable as is.

Essentially, unless we can compare machine reinforcement learning to human or animal intelligence, or we want to equate reinforcement learning to a cognitive process, then learning, as it relates to machines, can be seen as the computational goal of reinforcement learning as opposed to the "intelligent" learning humans do.

Question 2: Can you find a concrete empirical example or think of an example of your own to demonstrate Marr's three-level analysis, different from examples discussed in class (such as reinforcement learning, people waiting in a line etc.)? In your chosen example, can you show how much each level is independent and interdependent from one another? The example you choose may pertain to humans, animals, or machines. (It's fine if one level is not well defined or lacking in your example, identify it and explore potential reasons for its lacking.) (30%)

As we have learned, Marr's levels of analysis is helpful in understanding information processing systems like vision, our olfactory system, or computers. One particularly fascinating example of an information processing system that came up in class discussion last quarter (in Mind, Brain, and Meaning) was the record player. Although a record player is significantly less complex than cognitive processes of the brain, which is often the subject of Marr's levels of analysis, it represents information and presents and manipulates that information, some key components of an information processing system. Marr's description of vision as an information processing system by saying that, "Vision is a process that produces from images of the external world a description that is useful to the viewer and not cluttered with irrelevant information" (Marr, 1976; Marr and Nishihara, 1978, pg 31). Analogously, a record player produces sounds from an external object, vinyls, in such a way that the produced output is recognizable as the song the listeners intended to play from their chosen vinyl record.

Marr says that a machine can be seen as an information processing system if it can be understood on the computational, representational/algorithmic, and implementation levels of analysis (Marr, 25). At the computational level, we ask what is the goal of the computation and what is the logic behind being able to carry out that computation. To explore this, we will discuss the record player's inputs, outputs, and constraints. A record player's overall goal, as previously mentioned, is to produce sound, often music, that was encoded into a vinyl record - a record player… plays a record. The machine's inputs are the grooves on the vinyl, its output is sound, and some of its constraints are the duration of the song/size of the record, the health of the machine (is the needle broken), the speed at which the record wants to be played, and the depth and health of the grooves. While these constraints determine the output of a record player's computation, what is really important, "the nature of the problem", at the computational level is the production of sound (Marr, 27). We have already hinted at how the algorithmic and implementational levels impact the computational and vice versa. The success of the computation is largely dependent on the hardware of the record player and the production of music is only possible if we have an algorithm that can transform the grooves of a record into human hearable sound.

At the algorithmic level, we ask, what is the algorithm for the computational result we require? Already, I am asking a question that directly follows off of the computational level, highlighting the possible interplay of the two levels. It is important to note that if I explored the algorithmic level first, I could likely explore the computational level of the algorithm - why are we performing the algorithm in the first place? The implementation of the computational theory requires "transformations" of the input to output which a record player does by processing the representations of sound, grooves, into vibrations that are converted into electrical signals which are fed to amplifiers that play the sound. Again, this algorithm can only be implemented if the proper hardware is given - without an amplifier that can successfully amplify sound for human ears to hear, the goal of the computation will not be reached and the algorithm will be cut short.

Finally, on the implementational level, we think about the physical realization of the computation and algorithm. For a record player, we use a turning table to spin the record so that the music can be chronologically played. The wiggles in a vinyl record make the needle wiggle which changes the electric current in a magnet that the needle is connected to. These “wiggles” are the physical representation of sound that make the computation and algorithms of a record player actually possible. This specific implementation is really what makes a record player a record player, as opposed to a gramophone, although the two machines may have the same computational goal. The example of the gramophone also highlights the independence of the implementational level - the computation and algorithm do not *strictly* define the implementation. Overall, while the physical implementation, algorithm, and computational goal of a record player are all somewhat distinct, they all affect each other: it is the physical constraints of the device (implementation) which allow us to produce sounds from records (computational goal). It is the algorithm of creating and representation of sound, grooves and vibrations, which allows the computation to be performed. And, finally, the goal of the computational level and necessities dictated by each step in the algorithm which serve as a guiding force for the physical implementation of the record player.

Question 3: In your view, which approach more accurately assesses the capabilities of Large Language Models (LLMs): comparing LLMs' architecture and function to the human brain architecture and function, or evaluating their performance in the Turing test? Explain your reasoning and cite proper literature to support your argument (30%).

In 1950, Alan Turing introduced the Turing Test, initially called The Imitation Game”, as a test of machine/computer intelligence. Essentially, the test asked the question, could a computer successfully trick a human into believing they were conversing with another human? With the invention of LLM based systems like chat-GPT, heypi, or other chatbots, the Turing test is very relevant, especially in discussions surrounding how to build ethical AI systems. To me, the Turing test seems helpful as a benchmark - it is likely societally dangerous to create LLM based chatbots that are indistinguishable from humans (a point that Jones and Bergen echo in their paper’s abstract). But, if LLM behavior becomes indistinguishable from human behavior and language construction, then what that, more importantly, tells us is that, somehow, we have found a digital way to model language, a complex area of study for neurologists, psychologists, and computer scientists. As we become increasingly interested in assessing the capabilities of LLMs, we should compare their architecture and function to human brain architecture and function - the Turing test is merely one, although debated and criticized, method of doing so.

Searle's 1980 work, "Minds, Brains, and Programs" introduces the idea of Strong AI - the idea that artificial intelligence can have a mind or consciousness. Searle elaborates: "In strong AI, because the programmed computer has cognitive states, the programs are not mere tools that enable us to test psychological explanations; rather, the programs are themselves the explanations" (Searle, 417). While I will not touch on the validity of the distinction or overall idea of strong AI vs. weak AI, I think Searle is overall to the idea that the behavior of a computer program, AI, neural network, etc, has the ability to give us insight into how the brain works. If we can successfully program a computer to model or exhibit human intelligence, then it seems likely that through the process of creating that model, or in this case, LLMs, we have learned about or capitalized on some aspect of human behavior and function. At this point, I find myself asking the question, why create AI in the first place? What is it good for? Computer programs, historically, have helped humans solve human-solvable problems more efficiently - we can code solutions to problems we know the steps to, even if it is very difficult to execute those computations. It is clear, though, that LLMs are not regular computer programs. LLMs are informed by human language processes (hence the field of natural language processing) and their goal is to construct and manipulate language similar to humans.

Assessing LLM's on their capability to model human behavior and function also feels much more productive than simply analyzing it against the Turing test. In Jones and Bergen's evolution of GPT-4 in a public online Turing test, they found that GPT-4 did not pass the Turing test. They concede that, "Despite known limitations as a test of intelligence, we argue that the Turing Test continues to be relevant as an assessment of naturalistic communication and deception" (Jones & Bergen, 1). What I would focus on here, is that they see the Turing test, more importantly, as an evaluation for naturalistic communication, rather than intelligence. GPT-4 may not have passed the Turing test, but it can hold conversations with humans and deceive a significant number of interrogators in the imitation game. Just because GPT-4 failed the Turing test overall does not mean it has done great work in replicating human natural language processes. The study also found that interrogators in the study were influenced by the idea that they may be talking to a computer. This effect allowed the researchers to identify certain tactics and human characteristics that interrogators were looking for. This kind of data, though not entirely evidence based or scientific (due to the public and crowdsourced qualitative nature of the data) gives researchers an idea of what people define human intelligence or communication as. Overall, LLMs serve as a tool for understanding and replicating a vital human cognitive process, language, and should be evaluated on how well they do that. While the Turing test can help us test an LLM's proficiency, our overall goal is not to deem LLMs intelligent or not, and therefore, the Turing test should not be our sole evaluator.

Works Cited

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